

Extracting Implied Dividends from Options: An Analysis of The Coca Cola Company

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Introduction to The Coca-Cola Company

The Coca-Cola Company (NYSE: KO) is a multinational beverage corporation incorporated under the Delaware General Corporation Law and headquartered in Atlanta, Georgia. It is globally recognized for its flagship product, Coca-Cola, and operates a vast portfolio of non-alcoholic beverage brands [1].



From an investment perspective, Coca-Cola is highly regarded for its robust and reliable dividend history. The company is classified as a "Dividend King," a prestigious designation given to companies that have increased their annual dividend payouts for 50 or more consecutive years. In fact, Coca-Cola has increased its dividend for 64 consecutive years [2] [3]. As of recent data, the company

pays a quarterly dividend, with an annualized yield typically ranging between 2.7% and 3.0%, making it a staple in income-focused investment portfolios [4].

Methodology: The Put-Call Parity and Option Selection

To extract the implied dividend, we rely on the fundamental relationship between European call and put options known as the Put-Call parity. For a dividend-paying stock, the parity equation is expressed as:

$$C_0 - P_0 = S_0 - K \cdot D(0,T) - \text{Div}$$

Where:

- C_0 and P_0 are the current prices of the Call and Put options with strike K .
- S_0 is the current spot price of the underlying asset.
- $D(0,T)$ is the discount factor for maturity T .
- Div represents the value of the dividends paid during the life of the option.

Difficulty

The Put-Call Parity formula holds for European options, while those of “The Coca-Cola Company” are American ones, as explained by the CBOE, the main American stock exchange where these options are physically traded: [“Stock and ETF options, in contrast, are American-style. They can be exercised at any time before expiration.”](#)

Computing $D(0,T)$

First Attempt (sheet 1)

Initially, as suggested, I attempted to deduce the discount factor using wide strikes. We selected $K1 = 60$ and $K2 = 90$ for a 3-month maturity ($T=0.25$). The calculation yielded a discount factor $D(0,T) \approx 1.0007$. A discount factor greater than 1 implies negative interest rates, which is unrealistic in the current macroeconomic environment. Consequently, using this discount factor resulted in negative implied dividends, which is economically impossible for Coca-Cola. The error stems from the fact that deeply In-the-Money (ITM) options are the most sensitive to the **Early Exercise Premium**.

Second Attempt (Sheet 1)

I tried to avoid negative dividends by exploiting

<https://www.sitmo.com/extracting-high-precision-implied-dividend-and-interest-rate-term-structures-from-option-prices/>. Again the result was a discount factor greater than one, which confirms that we are dealing with an American option. It seems like the model compensates the inflated American prices by producing an impossible discount factor.

Sheet 1 — Initial OTM Attempt: Discount Factor Calculation (T = 3 months)

Strike	Put P_0	Call C_0	$C - P$
$K_1 = 60$	0.540	15.600	15.060
$K_2 = 70$	1.560	6.750	5.190
$K_3 = 72.5$	2.350	4.950	2.600
$K_4 = 75$	3.325	3.475	0.150
$K_5 = 77.5$	4.675	2.260	-2.415
$K_6 = 80$	6.250	1.365	-4.885
$K_7 = 90$	15.175	0.215	-14.960

Highlighted rows ($K_1=60$, $K_2=90$): initial OTM pair chosen. Discount Factor = $(C_{K1}-C_{K2}+P_{K2}-P_{K1})/(K_2-K_1) \approx 1.0007 > 1 \rightarrow$ unrealistic.

Possible Solution-Third attempt (sheet 1)

To resolve this issue, we adjusted our approach based on quantitative finance literature. We selected strikes that are **At-the-Money (ATM)**, specifically $K_1 = 72.5$ and $K_2 = 75$.

ATM American options have very little intrinsic value (in the sense of exercising the option immediately) and consist almost entirely of time value (the "Wait" value is higher or similar to the "Stop" value, referring to the lectures of Prof. Callegaro). Because the probability of early exercise for ATM options is negligible compared to deep ITM options, they behave much more like European options. Therefore, the Put-Call parity holds approximately true for ATM American options.

ATM Strike Selection (T = 3 months)

	$K_1 = 72.5$	$K_2 = 75$
Put P_0	2.350	3.325
Call C_0	4.950	3.475

$S_0 = 74.75$ | $T = 0.25$ yr

Discount Factor $D(0,T) = (C_{K1}-C_{K2}+P_{K2}-P_{K1})/(K_2-K_1)$
 $D(0,T) = (4.95-3.475+3.325-2.35)/(75-72.5) = 0.9800$

Derived Quantities (T = 3 months)

Quantity	Value
Discount Factor $D(0,T)$	0.9800
Forward Price F_0	75.153
Current Price S_0	74.750
Implied Dividend Div	1.122

$F_0 = (C_0-P_0)/D + K_1$ | $Div = S_0/D - F_0$

By using the ATM strikes, the recalculated discount factor for the 3-month maturity was $D(0,T) = 0.9800$. This is a realistic value (less than 1), reflecting a positive implied interest rate. With this valid discount factor, we successfully extracted a positive implied dividend of \$1.122.

Term Structure of implied dividends

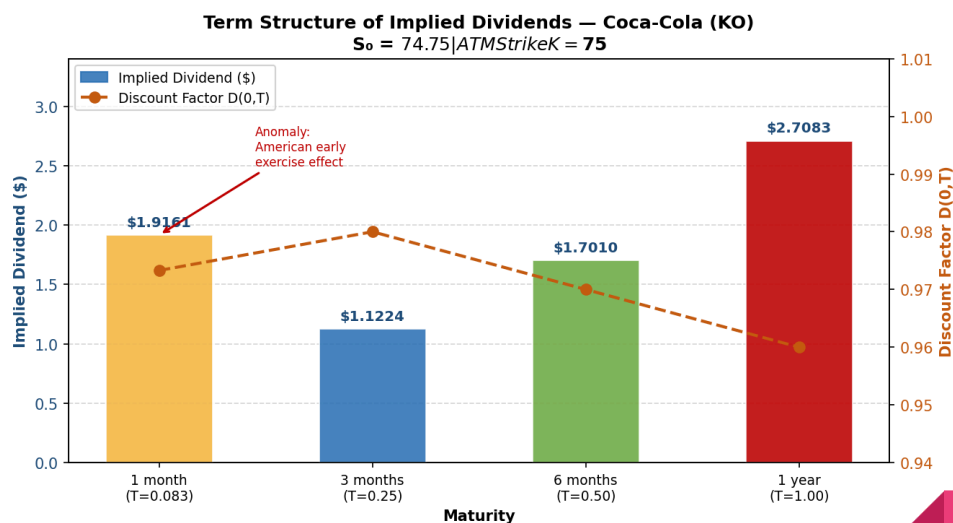
Having validated the ATM methodology, we expanded the analysis across multiple maturities to construct the term structure of implied dividends (as detailed in *Foglio 2*). We maintained the ATM strike $K = 75$ for the calculation of the forward price and implied dividend across all time horizons. The table below summarizes the extracted parameters for maturities of 1 month, 3 months, 6 months, and 1 year.

Sheet 2 — Full Term Structure: Implied Dividends across All Maturities

Maturity	T (years)	K_1	K_2	$C(K_1)$	$C(K_2)$	$P(K_1)$	$P(K_2)$	$D(0,T)$	F_0	Div (implied)
1 month	0.0833	73.0	76.0	3.275	1.590	1.205	2.440	0.9733	74.882	1.916
3 months	0.2500	72.5	75.0	4.950	3.475	2.350	3.325	0.9800	75.153	1.122
6 months	0.5000	72.5	75.0	6.350	3.575	4.875	4.525	0.9700	75.361	1.701
1 year	1.0000	77.5	80.0	5.700	4.650	7.250	8.600	0.9600	75.156	2.708

$S_0 = 74.75$ | ATM strike $K = 75$ used for all maturities | Anomaly noted at 1-month maturity (early exercise effect)

The extracted implied dividends are plotted in the chart below alongside the corresponding discount factors.



The term structure of implied dividends reveals several key insights:

1. **The 1-Month Anomaly:** The implied dividend for the 1-month maturity (**\$1.916**) appears anomalously high compared to the 3-month maturity (**\$1.122**). This is a known phenomenon in the pricing of short-term American options on dividend-paying stocks. As expiration approaches, the early exercise premium becomes highly sensitive to the upcoming ex-dividend date. Even ATM options can exhibit slight pricing distortions in the very short term due to the discrete nature of dividend jumps, causing the Put-Call parity to slightly misestimate the implied dividend.
2. **Upward Sloping Curve (3m to 1yr):** Excluding the 1-month anomaly, the term structure from 3 months to 1 year is upward sloping, progressing from \$1.122 to \$1.701, and reaching \$2.708 at the 1-year mark. This reflects the cumulative nature of dividends. A 1-year option encompasses four quarterly dividend payments, whereas a 3-month option encompasses only one.
3. **Discount Factor Consistency:** The discount factors exhibit a logical, monotonic decrease as maturity increases (from 0.9800 at 3 months down to 0.9600 at 1 year). This consistent downward trend confirms that the ATM methodology successfully neutralized the American option distortion across the longer maturities, providing a reliable yield curve estimation.

Conclusions

The \$2,7 estimate for the annual dividend is quite accurate with respect to the official \$2,17 one referring to [4]. The "synthetic" result represents the total cash expected plus the **Early Exercise Premium** accumulated over four dividend cycles. The resulting term structure accurately reflects the cumulative expectation of Coca-Cola's robust dividend payouts over time, save for minor short-term anomalies inherent to American options approaching expiration.

By shifting our focus to At-the-Money (ATM) options, which behave similarly to European options, we restored the validity of the Put-Call parity. This adjustment allowed us to compute realistic discount factors and successfully extract the implied dividends for Coca-Cola across various maturities. **However** with European options it is better to take $K_1 < K_2$ in order to avoid problematic noise generated by the Bid-Ask spread.

References

- [1] The Coca-Cola Company. (2026). *Company Overview*. Retrieved from <https://finance.yahoo.com/news/coca-cola-company-ko-dividend-032521810.html>
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- [3] Sure Dividend. (n.d.). *Dividend Aristocrats In Focus: Coca-Cola*. Retrieved from <https://www.suredividend.com/dividend-aristocrats-ko/>
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